CSC-580 – Advanced Software Engineering

Fall 2025 – Four Credit Hours

Instructor: John Hart
Course Location: MSB 104 & Zoom Classroom
Course Times (in-person): Tuesday - 4:30 PM – 6:00 PM via Zoom – REGISTER
Virtual Office Hours: Thursday - 5:00 PM to 6:00 PM [Optional] (Zoom Classroom)
Learning Management System: Canvas
Email: johnha@umich.edu
Phone: (517) 599-4829 (Text Preferred)

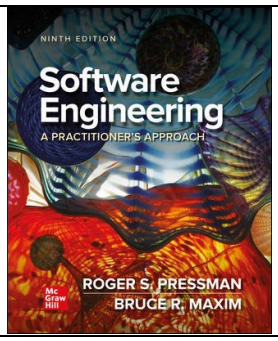
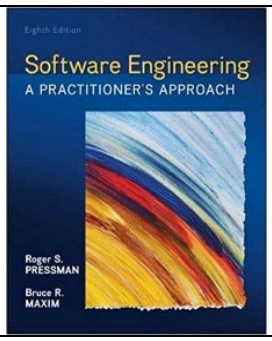
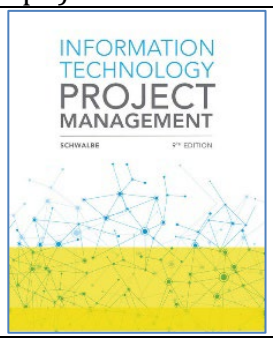
Prerequisites

Admission to the MS in CSIS program or instructor-approved override.

Course Description

Advanced topics in software engineering; advanced design methods including formal methods, component-based design, design with patterns and frameworks, and architectural-based designs. Modern software processes such as Extreme Programming and Agile software development. Issues and problems associated with large-scale software project failures and techniques for preventing them. Additionally, we will be examining the project management methodology for software development and how utilizing a strong mix of SE and PM practices will assist in project success.

Textbook

Software Engineering- A Practitioner's Approach by Pressman, 9th edition (Recommended – SE opt. 1)	*Software Engineering- A Practitioner's Approach* by Pressman, 8th edition (Recommended – SE opt. 2)	*Information Technology Project Management* by Schwalbe, 9th edition (Recommended PM opt.)
		

Other Good Reference Books (likely referenced in discussion):

- *Software Engineering* by Sommerville
- *Object-oriented Software Engineering* by Jacobson
- *Use Cases – requirements in context* by Kulak and Guiney
- *Writing Effective Use Cases* by Cockburn
- *Using CRC Cards* by Wilkinson
- *The Object Primer* by Ambler
- *Design Patterns Explained* by Shallow and Trott
- *Software Architecture* by Shaw and Garlan

Course Tools

Software Engineering Tools:

- *Microsoft Visio* - (drawing tool) Edu license available through Azure Edu
- *Draw.IO* – Freeware, web-based diagramming tool available here
- *Visual Paradigm* - Evaluation licenses available – Community Edition licenses free of charge (available at Visual Paradigm)
- *Visible Analyst* Student Edition (available at Visible Analyst for $49/term)

Project Management Tools:

- *Microsoft Project* - (project management scheduling and forecasting tool) Educational license available through Azure Edu

Course Objectives

The course objective is to provide you with the management techniques and skills required, to be an effective software engineer and future SE manager. As a future engineering professional, you have a responsibility to know what the latest industry standards, and software engineering processes, are and how they can be successfully applied. The objectives of this course are as follows:

- Provide understanding of software engineering processes.
- Enable you to clearly articulate the key issues associated with managing a software project.
- Give you an in-depth understanding of Object Modeling and UML
- Explore Component-based Software Engineering
- Provide you with a detailed understanding of Design Patterns
- Clearly understand the role of architecture in software design.
- Introduction to Formal Methods in Software Engineering
- Provide you with an understanding of software project management methodologies and approaches.

Course Format

The class will provide an active learning environment for everyone. Most in-person class sessions will consist of a traditional lecture, as well as real-time, interactive discussions. The second type of sessions (on-line - Zoom) will be used to break out and work in groups to reinforce the concepts covered in the lecture portion of the class period. Additionally, there will be supplemental (previously recorded) project management-related lectures that may be used to augment the student's learning. Since the instructor's desire is to grow student's skills in Software Engineering and grow their Project Management knowledge, the student is encouraged to partake in the additional content, meant to grow their project management knowledge and skill.

Class Location

This course is being offered in a hybrid modality – the primary lecture will be held on campus with synchronous Zoom session (recorded) offered for distance learners. And a secondary session which will be held fully online; delivered via Zoom.

Please register and ensure your UMich login is functional – both must be completed before you will be granted entry into the Zoom cyber classroom for lecture.

Cyber classroom: Zoom will be our virtual classroom – Registration Required!!!

Please REGISTER HERE – ***Be sure that your browser is signed in using your UMich Account!

After registering, you will receive a confirmation email containing information about joining each of the meetings.

***Note: you must also log into Umich.zoom.us with your UMich Credentials before you may enter the cyber classroom.

1) Zoom Classroom URL: Zoom UMich
2) Click Sign In (use UMich credentials)
3) Join Class Session (for that date)

Late Homework Policy

There will be ***NO*** late deliverables accepted - submissions are due by end of class. Once class ends, the Canvas submission link will close, and late deliverables will not be accepted.

Grading

- Project Management Assignments, Breakouts, Midterm and Final – combine to 500 points total for 100%:
  - PM Assignments: **240 pts**, Breakout: **60 pts**, Midterm: **100 pts**, and Final: **100 pts**. – **All Deliverables Below:** Groups Highly Encouraged!
    - PM Assignment is a series of seven graded deliverables (1-6 are mandatory; PM assignment 7 will be used as extra credit points [up to 5 points]).
    - Breakout Assignments is a series of seven graded deliverables (1-7 are mandatory; breakout 8 will be used as make-up points only) each worth 8.6 points.
    - Exams will be lengthy and will be worth 100 points each – please plan to dedicate 20-30 hours to each exam.
    - Each student will be pursuing 495+ points to achieve an A+ in the course.

Grading Scale:

A's	**B's**	**C's**	**D's and E**
99-100: A+	87-89: B+	75-79: C+	65-69: D+
92-98.9: A	82-86.9: B	72-74.9: C	62-64.9: D
90-91.9: A-	80-81.9: B-	70-71.9: C-	60-61.9: D- | 59.9-0: E

Narrative Rubric

Essay Grading Rubric		
Area of Concern	**Levels of Performance**	**Points Awarded**
Grammar and Organization	Answers are easy to read.	1
	Answers are difficult to read due to grammatical errors and/or lack of organization.	0
Completeness	Answers should leave no 'why questions' and/or missing technical content.	2
	Answers leave some questions and/or some missing technical content.	1
	Question is not fully answered.	0
Accuracy	All statements are accurate.	2
	Some statements are inaccurate.	1
	Mostly or completely inaccurate.	0

Drawing Rubric

Diagram Grading Rubric (5 points possible)		
Area of Concern	**Levels of Performance**	**Points Awarded**
Diagram Organization	Diagrams are professionally drawn.	1
	Diagrams are sloppily drawn.	0
Completeness	Diagram fully captures the essence of the problem.	2
	Diagrams are missing the essence of the problem.	1
	Diagrams are missing considerable portions of the essence of the problem.	0
Accuracy	All structural aspects of the diagrams accurate.	2
	Some structural aspects of the diagrams are inaccurate.	1
	Structural aspects of the diagrams are mostly or completely inaccurate.	0

Attendance

Regular class attendance is highly recommended unless students have informed the instructor that they are an approved distance learner. All students gain higher learning experiences when they participate in live lectures and breakout sessions. Because of this, points will be given to teams for breaking out and completing an exercise together. It is recommended that the break-out exercise be completed during the class breakout session, but teams can meet virtually at their convenience if they are made up of traditional or distance learners.

Note: International students in the U.S. may not take the class as distance learners due to U.S. government F-1 visa requirements.

Academic Misconduct

- Copying others' work, plagiarizing external references without giving due credit, and cheating in exams are strictly forbidden. The following is extracted from the UM-Flint Course Catalog:
- Intellectual integrity is the most fundamental value of an academic community. Students and faculty alike are expected to uphold the highest standards of honesty and integrity in their scholarship. No departure from the highest standards of intellectual integrity, whether by cheating, plagiarism, fabrication, falsification, or aiding and abetting dishonesty by another person, can be tolerated in a community of scholars. ***Such***

transgressions may result in actions ranging from: reduced grade, or failure of a course, to expulsion from the University, or revocation of degree.

- Please take pride in your own work, and feel free to ask me if you need help or assistance to maximize your learning. As shown above, the consequences of not doing your own work can be severe.
- If you have questions about what constitutes a violation of the University of Michigan's policies on Academic Integrity, please visit this page or contact me directly – I am happy to discuss (and explain) with any student.

Sexual Misconduct

The University of Michigan-Flint is committed to preventing sexual and gender-based misconduct and offering support to those who have been harmed. Sexual assault, harassment, discrimination, and all forms of sexual and gender-based misconduct have no place here. For more information, or to make a report, please visit the Equity, Civil Rights and Title IX Office (ECRT) at https://www.umflint.edu/ecrt.

Drugs and Alcohol

It is important for university employees and students to abide by the University of Michigan-Flint's Alcohol and Other Drugs (AOD) policy. It is your responsibility to familiarize yourself with the policy which is located here.

Accessibility and Disability Statement (and Accommodations)

The University of Michigan–Flint strives to make learning experiences as accessible as possible and complies with Section 504 of the Rehabilitation Act of 1973 and the American with Disabilities Act. The university provides individuals with disabilities, reasonable accommodation, to participate in educational programs, activities, and services. Students with disabilities requiring accommodations to participate in class activities or meet course requirements must self-identify with Disability and Accessibility Support Services as early as possible at (810) 762-3456 or dassflint@umich.edu. The office is located at 264 University Center, inside the CAPS Office. Once your eligibility for accommodation has been determined you will be issued an Accommodation Letter. Please present this letter to each faculty member in each class at the beginning of the term, or at least two weeks prior to the accommodation need.

Planned Schedule (Note: this schedule is subject to change)

Week	Topics
Week 1 (8/26/25) Discussion/Office Hours Only (8/28/25)	Lecture: Class Introduction and Syllabus Review Read (outside of class): - Schwalbe: Chapter 1 – Intro to Project Mgmt.
Week 2 (9/2/25) Discussion/Office Hours Only (9/4/25)	Lecture: Introduction to Software Engineering Read (before class): - Pressman 8th: Chapters 1 and 2 - Pressman 9th: Chapter 1 Read (outside of class): - Schwalbe: Chapter 2 – Project Mgmt. and the IT Context
Week 3 (9/9/25) Discussion/Office Hours Only (9/11/25)	Lecture: Waterfall and Agile Software Development Methodologies Read (before class): - Pressman 8th: Chapters 4 and 5 - Pressman 9th: Chapter 2, 3 and 4 Read (outside of class): - Schwalbe: Chapter 3 – Project Mgmt. Process Groups
Week 4 (9/16/25) Breakout & Discussion (9/18/25	Lecture: Systems Engineering & Business Process Models (Zachman) Read (before class): - Zachman (critical) - Pressman 8th: Chapter 7 - Pressman 9th: Chapters 5 and 6

	Read (outside of class): - Schwalbe: Chapter 4 – Project Integration Mgmt. Breakout Session 1: Scope section of Zachman Homework Assigned: PM part 1 (due no later than: 11/25/25 @ 11:59PM)
Week 5 (9/23/25) Breakout & Discussion (9/25/25)	Lecture: Object-oriented Analysis I – UML Overview & Use Cases Homework Due: PM part 1 (Recommended) Breakout Due: Breakout Session by 11:59PM Read (before class): - Pressman 8th: Chapter 8 - Pressman 9th: Chapter 7 Read (outside of class): - Schwalbe: Chapter 5 – Project Scope Mgmt. Breakout Session 2: Use Case Diagram and Text Use Case Homework Assigned: PM part 2 (due no later than: 11/25/25 @ 11:59PM)
Week 6 (9/30/25 Breakout & Discussion (10/2/25)	Lecture: Object-oriented Analysis II – Activity Diagrams, Swim Lane Diagrams & CRC Cards Homework Due: PM part 2 (Recommended) Breakout Due: Breakout Session 2 by 11:59PM Read (before class): - Pressman 8th: Chapter 9 - Pressman 9th: Chapter 8 (8.3 and 8.5) Read (outside of class): - Schwalbe: Chapter 6 – Project Schedule Mgmt. Breakout Session 3: Activity Diagrams, Swim Lanes & CRC

	Cards
Week 7 (10/7/25) Breakout & Discussion (10/9/25)	Lecture: Object-oriented Analysis III – Class Diagrams & Associations, Aggregations, Dependencies Breakout Due: Breakout Session 3 by 11:59PM Read (before class): - Pressman 8th: Chapter 10 - Pressman 9th: Chapter 7 and 8 Read (outside of class): - Schwalbe: Chapter 7 – Project Cost Mgmt. Breakout Session 4: Associations, Aggregations, Dependencies Homework Assigned: PM part 3 (due no later than: 11/25/25 @ 11:59PM)
Week 8 (10/14/25) Breakout & Discussion (10/16/25)	Lecture: Object-oriented Analysis IV – Sequence Diagrams & State Diagrams Homework Due: PM part 3 (Recommended) Breakout Due: Breakout Session 4 by 11:59PM Read (before class): - Pressman 8th: Chapter 11 - Pressman 9th: Chapter 8 (8.4 and 8.5) Read (outside of class): - Schwalbe: Chapter 8 – Project Quality Mgmt. Breakout Session 5: Sequence Diagrams & State Diagrams Homework Assigned: PM part 4 (due no later than: 11/25/25 @ 11:59PM)
Week 9 (10/21/25 and 10/23/25)	No Classes Held – Fall Break Midterm Exam (take home) – Assigned: 10/14/25 @ 7:00PM |

No Class and No Breakout or Discussion Session	Due: 10/24/25 by 11:59PM
Week 10 (10/28/25) Breakout or Discussion Session (10/30/25)	Lecture: Software Architecture I Breakout Due: Breakout Session 5 Homework Due: PM part 4 (Recommended) Read (before class): - Pressman 8th: 13 - Pressman 9th: 10 Read (outside of class): - Schwalbe: Chapter 9 – Project Resource Mgmt. Breakout Session 6: Non-functional Req. Evaluation and Rating
Week 11 (11/4/25) Breakout & Discussion (11/6/25)	Lecture: Software Architecture II Breakout Due: Breakout Session 6 by 11:59PM Read (before class): - Pressman 8th: 13 - Pressman 9th: 10 Read (outside of class): - Schwalbe: Chapter 10 – Project Communications Mgmt. Breakout Session 7: Architecture Evaluation and Selection Homework Assigned: PM part 5 (due no later than: 11/25/25 @ 11:59PM)
Week 12 (11/11/25)	Lecture: Designing for Reuse & Component-based Development Homework Due: PM part 5 (Recommended)

No Breakout or Discussion Session (11/13/25)	Breakout Due: Breakout Session 7 by 11:59PM Read (before class): - Pressman 8th: Chapter 14 - Pressman 9th: Chapter 11 Read (outside of class): - Schwalbe: Chapter 11 – Project Risk Mgmt. Homework Assigned: PM part 6 (due no later than: 11/25/25 @ 11:59PM)
Week 13 (11/18/25) Breakout & Discussion (11/20/25)	Lecture: Frameworks and Design Patterns Homework Due: PM part 6 (Recommended) Read (before class): - Pressman 8th: Chapter 16 - Pressman 9th: Chapter 14 Read (outside of class): - Schwalbe: Chapter 12 – Project Procurement Mgmt. Breakout Session 8: Framework Evaluation and Selection (Non-functional Requirements Copied from Architecture Selection) Homework Assigned: PM part 7 (optional) (due no later than: 11/25/25 @ 11:59PM)
Week 14 (11/25/25) No Breakout or Discussion Session (11/27/25)	Lecture: Software Testing Homework Due: PM part 7 (optional) Breakout Due: Breakout Session 8 by 11:59PM (optional) Read (before class): - Pressman 8th: Chapters 22, 23, and 24 - Pressman 9th: Chapters 19 and 20

	Read (outside of class): - Schwalbe: Chapter 13 – Project Stakeholder Mgmt.
Week 15 Final Exam (12/2/25)	No Class Held – Final Exam Study Day Final Exam (take home) – Assigned: 11/25/25 @ 7:00PM | Due: 12/5/25 by 11:59PM